Sahaj Bhadja

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Summary

Software engineer with 2+ years building **distributed backends**, **secure APIs**, and **cloud workflows** for reliable, blockchain-oriented infrastructure.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Architected **Azure SQL and Blob services** for 500+ users, tuning indexed queries to cut dashboard latency from **1.2 seconds to 350 milliseconds**.
- Built **fault-tolerant job execution** for 1,000+ monthly workflows with retries and isolation, improving throughput by **3x** under concurrent production workloads.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Engineered **Flask APIs** and React workflows handling **10,000+ daily requests**, integrating secure ingestion and automated grading for production education workflows.
- Designed **multithreaded processing** for 1,000+ uploads on AWS RDS-backed services, reducing grading turnaround by **60%** while improving throughput and debugging.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered **Python API and ETL microservices** with SQL, AWS Glue, and S3, improving response times by **30%** across distributed data workflows.
- Maintained **Jenkins CI pipelines** and 20+ automated suites with PyTest and Mockito, increasing release reliability by **68%** across shared environments.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built **REST APIs and WebSocket modules** supporting **2M+ synchronized records**, improving consistency across real-time modular workflows.
- Automated **Docker build and test workflows** with Jenkins CI/CD, cutting release time by **50%** while improving deployment consistency.

Pandit Deendayal Energy University **India**
Research Fellowship *January 2023 – May 2024*
- Designed a **quantum-resistant blockchain architecture** with 5+ lattice cryptographic primitives for decentralized protocol security.
- Benchmarked **consensus mechanisms** and proposed an approach reducing centralized-control and quantum-risk exposure by **25%**, contributing to a peer-reviewed IISc paper.

Skills

Languages: **TypeScript**, **Python**, JavaScript, SQL, Java, C++, Bash; **Backend**: REST APIs, FastAPI, Flask, Node.js, WebSockets, microservices, multithreading, concurrency
Data & Cloud: PostgreSQL, MySQL, SQL Server, Azure SQL, MongoDB, AWS, Azure, S3, Blob Storage, AWS Glue, ETL, indexing, query tuning; **Security**: blockchain, cryptography, consensus algorithms, decentralized systems
Delivery & Quality: Docker, Jenkins, CI/CD, Git, Kubernetes, PyTest, Mockito, automated testing, reliability engineering, incident debugging, Agile/Scrum

Projects

FinNexus Live | *Next.js, TypeScript, Kubernetes, Plaid*
- Built a scalable **fintech platform** for **50,000+ users** and **1M+ transactions**, combining secure APIs, Plaid, and real-time synchronization.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*
- Built an **AI-powered SQL workspace** for **10+ databases**, accelerating query authoring and schema exploration through unified Java and TypeScript services.

Blockchain Authentication Paper Code | *Blockchain, Authentication, Security Research*
- Designed a **decentralized authentication model** replacing centralized verification with tamper-proof login flows and role-based access across distributed applications.